β pleated sheets

In a β pleated sheet, two or more segments of a polypeptide chain line up next to each other forming a sheet like structure held together by hydrogen bonds. The strands of a β pleated sheet may be parallel where the N- and C- termini of the strands match up or antiparallel where the N-terminus of one strand is positioned next to the C-terminus of the other.

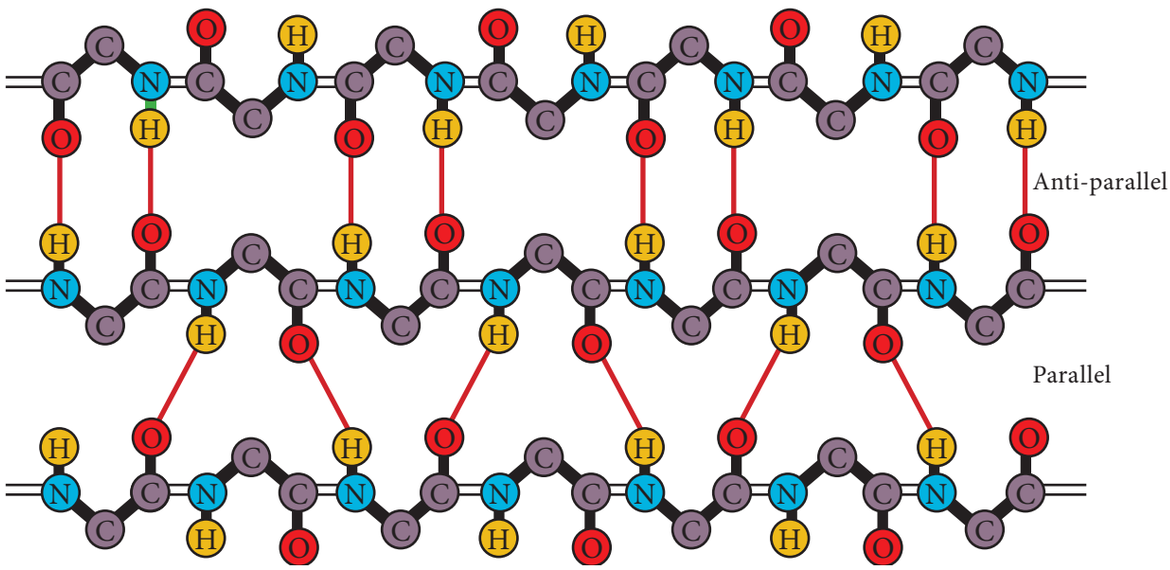


Figure 3.10 : structures of parallel and anti parallel β sheets

Table 3.2 comparison between α helix and β pleated sheet

α Helix	β Sheet
1. The interacting residues are always from a continuous stretch of polypeptide chain.	1. The interacting residues are not from a continuous stretch of a polypeptide chain.
2. Forms a tightly coiled structure	2. Forms a fully extended structure
3. The hydrogen bonds are parallel to direction of the polypeptide backbone	3. The hydrogen bonds are perpendicular to the direction of the polypeptide back bone
4. Can exist as a right handed or left handed helix	4. Can exist as parallel or anti parallel sheets
5. The amino acids Methionine, alanine, leucine, glutamic acid and lysine have higher probabilities of occurrence in α helix.	5. The amino acids isoleucine,valine,threonine, phenylalanine and tyrosine have higher probabilities of occurrence in a β sheet.

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